

Internet of Things

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Outline

- ▶ Wearable Technology and the Internet of Things
- ▶ Key Components of Wearable Devices
- ▶ Techniques and Technologies for Wearable Devices
- ▶ Applications of Wearable Devices
- ▶ Future improvement in Wearable Devices
- ▶ Practice code Wearable Devices IoT

Wearable Technology and the IoT

- ▶ Wearable technology encompasses electronic devices that can be worn on the body or attached to clothing, designed to offer convenience and often hands-free operation. These devices present a spectrum of complexity, from basic fitness trackers that monitor simple metrics like steps taken and heart rate, to more advanced gadgets capable of providing access to data, facilitating communication, or remotely controlling other devices. Typically powered by small, efficient batteries, wearables are engineered to be lightweight and discreet, seamlessly integrating into the user's daily life.



Wearable Technology and the IoT

- ▶ Initially considered niche gadgets appealing primarily to technology enthusiasts, wearable devices have undergone significant evolution to become indispensable tools across a multitude of industries, including healthcare, fitness, logistics, and retail. The trajectory of their development reveals a clear trend: an initial focus on fundamental tracking functionalities has expanded to incorporate increasingly sophisticated features and specialized applications. This evolution underscores the growing importance of wearables in both personal and professional contexts. For instance, early fitness trackers primarily counted steps. The integration of these devices with the Internet of Things has unlocked capabilities for more complex data collection, such as continuous heart rate monitoring and sleep analysis, and enabled interaction with other connected devices. This has paved the way for highly specialized applications in fields like healthcare, where wearables now play a crucial role in remote patient monitoring, and in industrial settings, where they enhance worker safety and efficiency. This progression signifies a maturing technology that is finding diverse and impactful uses across various sectors

Key Components of Wearable Devices

The functionality of wearable devices hinges on the intricate interplay of several key components: **sensors, processors, connectivity modules, and a power source.**

► Sensors are the data-gathering elements of wearables, tasked with collecting information from the user's body or the surrounding environment. A wide array of sensors can be integrated into these devices, depending on their intended purpose. **Motion tracking, for example, is often achieved using accelerometers and gyroscopes, while heart rate monitoring typically employs optical sensors.** Wearables can also include sensors to **analyze sleep patterns, measure body temperature, detect UV exposure,** or even assess air quality. For instance, a basic fitness tracker relies on an **accelerometer to count steps and might include a heart rate sensor to monitor cardiovascular activity.**

Key Components of Wearable Devices

- ▶ The collected raw data is then channeled to processors, often in the form of microcontrollers, which serve as the computational core of the wearable. These processors execute algorithms and software to transform the raw signals from the sensors into meaningful metrics and insights. Given the size and power constraints of wearable devices, microcontrollers known for their low power consumption and adequate processing capabilities, such as those in the ARM Cortex-M series, are commonly utilized.

Key Components of Wearable Devices

- ▶ To enable communication with other devices and the internet, wearables are equipped with connectivity modules. Various wireless technologies facilitate this data exchange, with Bluetooth and Bluetooth Low Energy (BLE) being prevalent for short-range communication with smartphones or dedicated hubs. Wi-Fi may be incorporated for broader network connectivity when power consumption is less of a concern or when the device is within range of a known network.

Key Components of Wearable Devices

- ▶ The power source for most wearables is a small, energy-efficient battery, typically of the lithium-polymer type, designed to maximize the device's operational time. Given the continuous nature of wearable use, extending battery life is a critical design consideration. Consequently, research and development efforts are increasingly exploring energy-harvesting technologies as a means to supplement battery power or even create self-powered wearables in the future. The selection of low-power components, such as ARM Cortex-M microcontrollers, and the implementation of smart power management integrated circuits are key strategies in controlling power distribution and ensuring efficient energy usage.

Key Components of Wearable Devices

- ▶ The seamless operation and extended usability of wearable devices depend significantly on the careful selection and integration of these components, particularly the **balance between functionality and power efficiency, given the inherent size and weight limitations.**

Examples of Common Wearable Devices

- ▶ **Fitness Trackers** are designed primarily to monitor an individual's health and activity levels. They typically track metrics such as the number of steps taken, calories burned, distance traveled, and sleep duration and quality. **Many also include heart rate monitoring capabilities. Popular examples of fitness trackers include products from Fitbit, Garmin, and Xiaomi Mi Band.**
- ▶ **Smartwatches** offer a broader range of functionalities compared to basic fitness trackers. While they often include activity and **health tracking features**, smartwatches also provide **notifications from smartphones** (such as calls, messages, and app alerts), **GPS navigation**, and some even **support contactless payments**. They essentially function as miniaturized computers worn on the wrist. Prominent examples in this category are the Apple Watch and the Samsung Galaxy Watch.

Examples of Common Wearable Devices

- ▶ **Smart Glasses** represent a more advanced form of wearable technology that integrates augmented reality (AR) capabilities. These devices can overlay digital information onto the user's field of vision, providing hands-free access to data, instructions, or real-time information. Applications range from industrial use, where they might **display repair manuals, to consumer applications for enhanced navigation or information retrieval.** Examples of smart glasses include Google Glass and Microsoft HoloLens.



Examples of Common Wearable Devices

- ▶ **Smart Clothing** integrates sensors directly into fabrics to monitor physiological data. These garments can track metrics like heart rate, body temperature, respiratory rate, and muscle activity. Applications are diverse, ranging from athletic performance monitoring to healthcare and even military uses.
- ▶ **Medical Wearables** are specifically designed for health monitoring and management. This category includes devices like continuous glucose monitoring systems used by individuals with diabetes (e.g., Dexcom, Eversense) and wearable defibrillators for patients at risk of cardiac arrest (e.g., Zoll LifeVest). These devices often provide critical, real-time health data to both users and healthcare professionals.

How Wearable Devices Fit into the IoT Framework

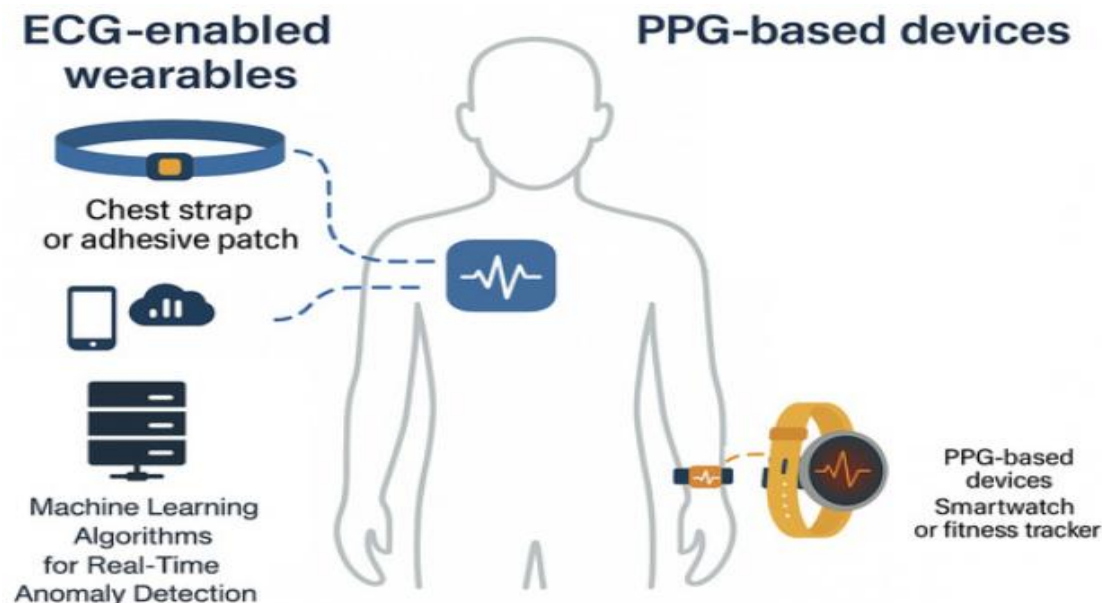
- ▶ Wearable devices designed to be worn on the body, serve as a direct and often continuous interface between individuals and the vast network of connected objects and services that constitute the IoT. By embedding sensors, processors, and connectivity capabilities, wearables seamlessly integrate into this ecosystem, acting as specialized endpoints that focus on collecting data about the user and their immediate environment. This personal data then becomes a valuable part of the extensive information flow within the IoT, contributing to a more personalized and context-aware connected experience

How Wearable Devices Fit into the IoT Framework

- ▶ Furthermore, wearables are not merely passive data collectors; they actively participate in the IoT framework by enabling users to interact with and control other connected devices and systems. For instance, a smartwatch can receive notifications from a multitude of IoT applications running on a smartphone or other connected devices, providing timely updates and alerts directly to the user's wrist. Additionally, many smartwatches can function as remote controls for smart home devices, allowing users to adjust settings for lighting, temperature, or even security systems with a simple tap or voice command. This capability positions wearables as key intermediaries within the IoT, simplifying interactions and making connected environments more intuitive and accessible.

How Wearable Devices Fit into the IoT Framework

- ▶ In essence, wearable devices extend the reach and **usability of the IoT by providing a personal, immediate, and often hands-free connection to the digital world.** They embody the core principles of the IoT by sensing, collecting, processing, and exchanging data, ultimately enhancing the user's interaction with and experience within the broader network of connected things.



The Role of Wearables as Data Collection Points in IoT

- ▶ A primary function of wearable devices within the Internet of Things (IoT) is their role as sophisticated data collection points, continuously gathering a diverse range of information about the user's physiological state, their physical activities, and the surrounding environment. Equipped with an array of embedded sensors, wearables are capable of monitoring a multitude of metrics, including vital signs such as heart rate and blood oxygen levels, patterns of physical activity like steps taken and calories burned, sleep duration and quality, body temperature fluctuations, and even environmental factors like exposure to ultraviolet (UV) radiation and the quality of the air.

The Role of Wearables as Data Collection Points in IoT

- ▶ This personal data, collected **unobtrusively** as users go about their daily routines, contributes a unique and exceptionally valuable layer to the vast data pool that constitutes the IoT. Unlike other types of IoT sensors that might monitor industrial processes or environmental conditions on a broader scale, **wearables focus on the individual, providing a rich and continuous stream of personal data**. This wealth of information enables the development of highly personalized applications and services tailored to the specific needs and preferences of the user, driving advancements in areas such as **precision healthcare, individualized fitness regimens, and proactive wellness management**.

The Role of Wearables as Data Collection Points in IoT

- ▶ The intimate and continuous nature of data collection by wearables makes them a cornerstone of the personal IoT experience, providing insights and enabling interventions that can significantly impact an individual's health, well-being, and interaction with their connected world.



Enabling Interaction and Control Through Wearable IoT

- ▶ Beyond their crucial role as data collection points, wearable devices also serve as a convenient and often intuitive means for **users to interact with and exert control over the broader ecosystem of connected objects and systems within the Internet of Things (IoT)**. Unlike traditional interfaces that might require users to physically manipulate a device like a smartphone or computer, wearables offer a more immediate **and accessible form of interaction**.

Enabling Interaction and Control Through Wearable IoT

- ▶ Smartwatches, for example, can receive and display notifications from a wide range of IoT applications, including those related to [smart home devices](#), [social media](#), and [productivity tools](#). This allows users to stay informed and manage their digital lives without having to constantly reach for their smartphones. Furthermore, many wearables are equipped with features that enable them to directly control other connected devices. [A smartwatch might be used to adjust the settings of a smart thermostat, turn smart lights on or off, or even manage music playback on a connected speaker system.](#)

Enabling Interaction and Control Through Wearable IoT

- ▶ In certain scenarios, wearables can even act as gateways, facilitating communication between different types of connected devices that might not otherwise be able to interact directly. This capability enhances the overall interoperability and user experience within a smart environment. The convenience and often hands-free nature of wearables make them an increasingly preferred interface for managing and interacting with the growing number of connected devices that permeate our daily lives, making the IoT more accessible and user-friendly.

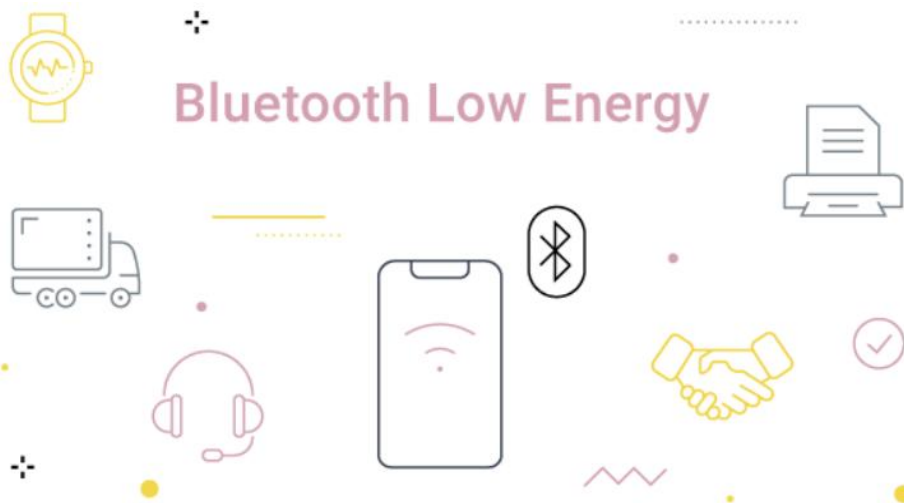
Communication Protocols in Wearable IoT

- ▶ **Short-Range Communication Protocols (Bluetooth, BLE, NFC)**
- ▶ Wearable devices rely heavily on short-range wireless communication protocols to interact with nearby devices, primarily smartphones, and to facilitate specific functionalities like **contactless payments**. Three of the most prevalent protocols in this context are **Bluetooth**, **Bluetooth Low Energy (BLE)**, and **Near Field Communication (NFC)**.
- ▶ Bluetooth is a widely adopted wireless technology that enables the exchange of data over relatively short distances, **typically up to 10 meters**. It **supports both voice and data transmission**, making it versatile for various applications in wearables, such as connecting to smartphones for **hands-free calls**, **streaming audio to wireless headphones**, and **syncing data with companion apps**. While Bluetooth offers a convenient way to establish connections and transfer data, it generally **consumes more power** compared to its low-energy counterpart.



Communication Protocols in Wearable IoT

- ▶ Bluetooth Low Energy (BLE), is a version of Bluetooth designed specifically for devices that require minimal power consumption, making it an ideal choice for battery-powered wearables. BLE maintains a similar communication range to classic Bluetooth but achieves significantly lower energy usage, which is crucial for extending the battery life of devices like fitness trackers and smartwatches that are intended for continuous wear. It is commonly used for transmitting health and fitness data between wearables and smartphones.



IoT Centric Design

BLE remains in sleep mode

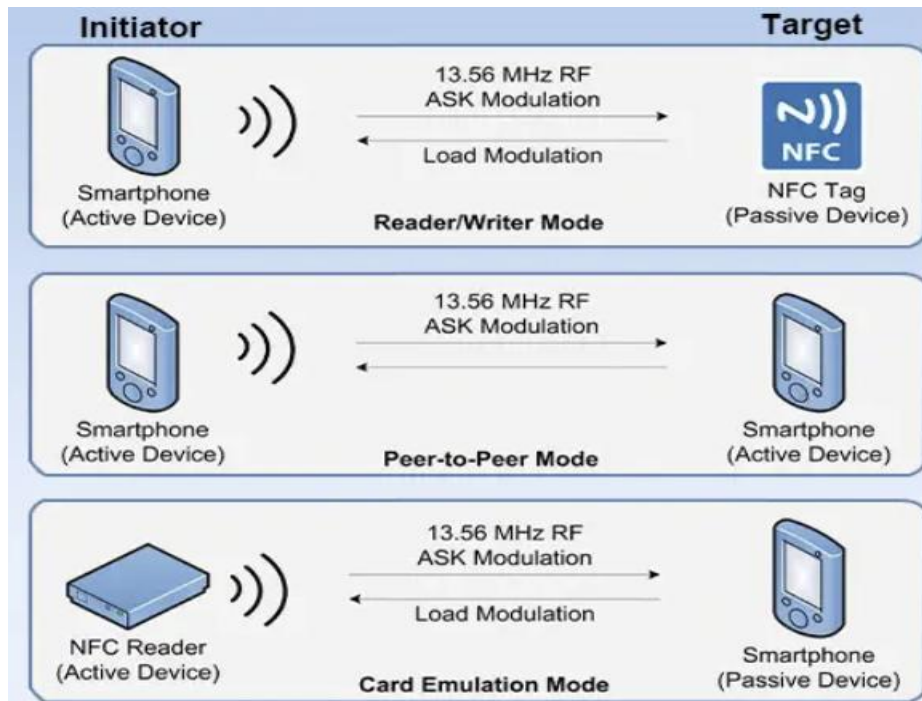
Communication Protocols in Wearable IoT

- ▶ **Near Field Communication (NFC)** is a communication protocol that enables two electronic devices to exchange data when they are brought within very **close proximity**, typically a distance of 4 centimeters or less.



Communication Protocols in Wearable IoT

- ▶ NFC offers a low-speed connection with a simple setup process and is particularly well-suited for applications requiring secure, short-range interactions, such as contactless payments made with smartwatches or fitness bands, accessing secure areas with smart rings, and quickly pairing wearables with other devices.



Data Flow - How Wearables Exchange Information within IoT

- ▶ **Data Collection from Wearable Sensors**
- ▶ Wearable devices are designed with a **diverse array of sensors**, each meticulously chosen to monitor specific aspects of the user's physiology and the surrounding environment. These sensors work continuously to gather real-time data, forming the initial and most critical step in the wearable IoT system.
- ▶ **Transmission of Data to Gateways or Smartphones**
- ▶ Once the sensors on a wearable device have gathered the necessary data, this information needs to be efficiently and reliably transmitted to other devices or platforms for further processing and analysis. **The primary means of achieving this data transmission is through short-range wireless communication protocols, with Bluetooth and, particularly, Bluetooth Low Energy (BLE) being the most commonly utilized technologies**

Data Flow - How Wearables Exchange Information within IoT

- ▶ **Data Transfer to Cloud Platforms for Processing and Analysis**
- ▶ The smartphone or IoT gateway, having successfully received the data from the wearable device via a short-range protocol, then takes on the crucial task of transferring this information to cloud-based IoT platforms. This transfer is typically facilitated by the smartphone or gateway's own internet connection, utilizing either a Wi-Fi network or a cellular data connection. Prominent cloud platforms such as AWS IoT, Azure IoT, and Google Cloud IoT are specifically designed to handle the massive influx of data generated by potentially millions of connected devices, including wearables.

Data Flow - How Wearables Exchange Information within IoT

- ▶ **Feedback and Control Signals from IoT Platforms to Wearables**
- ▶ The flow of information within a wearable IoT system is not unidirectional; it often involves a crucial feedback loop where IoT platforms send signals back to the wearable devices. **This bidirectional communication is essential for creating truly interactive and responsive systems.** The feedback can be based on the analysis of the data collected by the wearable, or it can be in direct response **to commands issued by the user through a connected application.**

Applications and Use Cases of Integrated Wearable IoT

- ▶ **Healthcare and Wellness Monitoring**
- ▶ **Fitness and Activity Tracking**
- ▶ **Smart Homes and Automation**
- ▶ **Industrial and Enterprise Applications**
- ▶ **Navigation and Location Services**
- ▶ **Accessibility and Assistance**

Industrial and Enterprise Applications

- ▶ Wearables are used to monitor the safety of workers in hazardous environments by tracking their location and detecting potential dangers like falls or exposure to harmful substances. Smart glasses and headsets provide hands-free access to critical information and instructions for field workers, improving efficiency and safety during complex tasks. In logistics and warehousing, wearable scanners and RFID readers streamline inventory management and asset tracking, leading to increased accuracy and reduced errors. Additionally, wearables can monitor worker fatigue and stress levels in safety-critical industries, helping to prevent accidents caused by exhaustion.

Navigation and Location Services

- ▶ Wearable devices with GPS capabilities offer convenient navigation assistance for various activities. Smartwatches and fitness trackers can provide turn-by-turn directions for walking, running, or cycling, eliminating the need to constantly check a smartphone. Wearables equipped with location tracking features are also used for personal safety, allowing parents to monitor the whereabouts of their children or caregivers to keep track of elderly individuals.

Accessibility and Assistance

- ▶ Wearable IoT technology holds immense potential for improving the lives of individuals with disabilities and accessibility needs. Smart gloves with integrated sensors can provide navigational cues through haptic feedback for visually impaired users, enabling them to navigate independently. Wearable devices with fall detection features can automatically alert emergency contacts or services if the wearer experiences a sudden fall, providing crucial assistance for the elderly or those with mobility issues. Furthermore, researchers are exploring the use of wearables as communication aids for individuals with hearing or speech impairments, with potential applications in sign language translation and text-to-speech conversion

Recent Trends and Updates in Wearable IoT

▶ **Advancements in Sensor Technology**

Researchers and developers are constantly innovating to create sensors that are not only smaller, more accurate, and more energy-efficient but also capable of tracking an ever-expanding range of physiological and environmental data. This includes the ongoing development of novel sensors that could enable advanced health monitoring features in wearables, such as continuous, non-invasive methods for tracking blood pressure and glucose levels, as well as the detection of various biomarkers in bodily fluids like sweat or interstitial fluid.

▶ **Improvements in Battery Life and Power Efficiency**

▶ **Enhanced Connectivity Options**

▶ **Integration of AI and Machine Learning in Wearable IoT**

▶ **Focus on Security and Privacy in Wearable IoT Ecosystems**

Practice code wearable IoT

- ▶ Extend the `WearableDevice` class to simulate more realistic sensor behavior.
- ▶ Add a `activity_level` property to the `WearableDevice` class (e.g., "sedentary," "walking," "running").
- ▶ Modify the `get_sensor_data()` method to make the heart rate and oxygen level vary based on the `activity_level`.
- ▶ When `activity_level` is "sedentary," the heart rate should have a small variation (e.g., between 60 and 80), and the oxygen level should be stable (e.g., between 95 and 100).
- ▶ When `activity_level` is "walking," the heart rate should increase moderately (e.g., between 80 and 120), and the oxygen level might decrease slightly.

Reference

- ▶ <https://www.minew.com/guide-of-iot-wearable/#:~:text=IoT%20wearables%20are%20a%20category,%2C%20often%20in%20real%2Dtime.>
- ▶ <https://www.hashstudioz.com/blog/iot-in-wearables/#:~:text=in%20Wearable%20Technology-,Defining%20IoT%20in%20Wearables,Wi%2DFi%2C%20Bluetooth%2C%20NFC>

Question



Thank you

